mammals and the flippers of Penguins and Dolphins. Root modification in sweet potato and stem modification in potato are considered as analogous organs. Both of these plants have a common function of storage of food.

Vestigial organs

Structures that are of no use to the possessor, and are not necessary for their existence are called vestigial organs. Vestigial organs may be considered as remnants of structures which were well developed and functional in the ancestors, but disappeared in course of evolution due to their non-utilization. Human appendix is the remnant of caecum which is functional in the digestive tract of herbivorous animals like rabbit. Cellulose digestion takes place in the caecum of these animals. Due to change in the diet containing less cellulose, caecum in human became functionless and is reduced to a vermiform appendix, which is vestigial. Other examples of vestigial organs in human beings include coccyx, wisdom teeth, ear muscles, body hair, mammae in male, nictitating membrane of the eye, etc.,

Connecting link

The organisms which possess the characters of two different groups (transitional stage) are called connecting links. Example *Peripatus* (connecting link between Annelida and Arthropoda), *Archeopteryx* (connecting link between Reptiles and Aves).

Atavistic organs

Sudden appearance of vestigial organs in highly evolved organisms is called atavistic organs. Example, presence of tail in a human baby is an atavistic organ.

6.4.3 Embryological evidences

Embryology deals with the study of the development of individual from the egg to the adult stage. A detailed study of the embryonic development of different forms makes us to think that there is a close resemblance during development.

The development of heart in all vertebrates follows the same pattern of development as a pair of tubular structures that later develop into two chambered heart in fishes, three chambered in amphibians and in most reptiles and four chambered in crocodiles, birds and mammals; indicating a common ancestry for all the vertebrates,

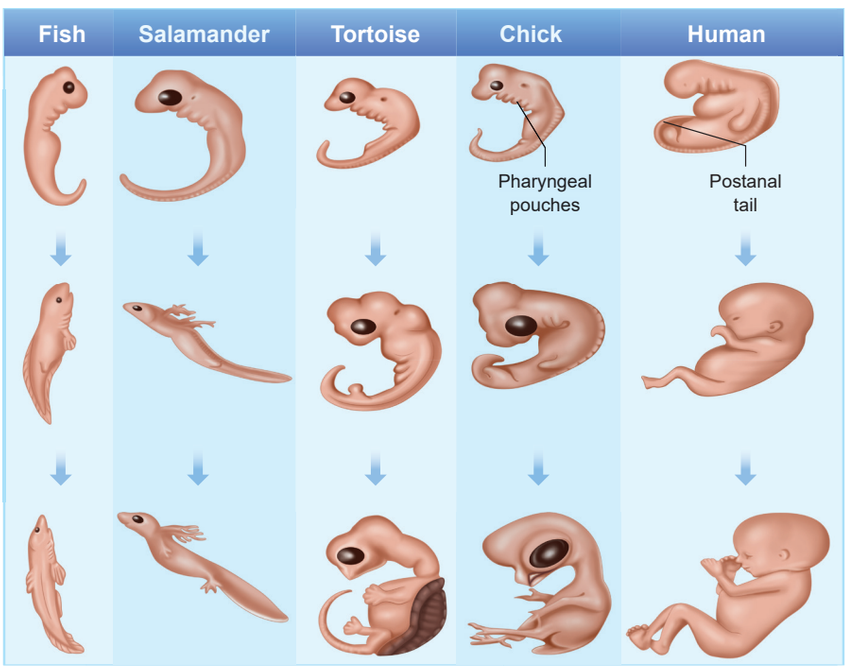


Fig 6.4 Embryological evidences

Hence scientists in the 19th century concluded that higher animals during their embryonic development pass through stages of lower animals (ancestors). **Ernst Von Haeckel**, propounded the **"biogenetic law** or

XII Std Zoology Chapter 6 EM.indd 98 23/12/2022 16:12:49